

```

< switched to db day2
> db.createCollection("patients")
< { ok: 1 }
> db.patients.insertMany([
  { "patientName": "Arun", "doctor": "Dr. Ravi", "department": "Cardiology", "age": 45, "treatmentCost": 20000, "daysAdmitted": 5 },
  { "patientName": "Meena", "doctor": "Dr. Kumar", "department": "Neurology", "age": 50, "treatmentCost": 30000, "daysAdmitted": 7 },
  { "patientName": "Raj", "doctor": "Dr. Ravi", "department": "Cardiology", "age": 60, "treatmentCost": 15000, "daysAdmitted": 3 },
  { "patientName": "Divya", "doctor": "Dr. Mehta", "department": "Orthopedics", "age": 35, "treatmentCost": 10000, "daysAdmitted": 2 },
  { "patientName": "Kiran", "doctor": "Dr. Kumar", "department": "Neurology", "age": 40, "treatmentCost": 25000, "daysAdmitted": 6 },
  { "patientName": "Sneha", "doctor": "Dr. Mehta", "department": "Orthopedics", "age": 30, "treatmentCost": 8000, "daysAdmitted": 2 },
  { "patientName": "Vijay", "doctor": "Dr. Ravi", "department": "Cardiology", "age": 55, "treatmentCost": 22000, "daysAdmitted": 4 },
  { "patientName": "Priya", "doctor": "Dr. Kumar", "department": "Neurology", "age": 48, "treatmentCost": 27000, "daysAdmitted": 5 },
  { "patientName": "Anil", "doctor": "Dr. Mehta", "department": "Orthopedics", "age": 52, "treatmentCost": 18000, "daysAdmitted": 4 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a115ef49060e076723cf781'),
    '1': ObjectId('6a115ef49060e076723cf782'),
    '2': ObjectId('6a115ef49060e076723cf783'),
    '3': ObjectId('6a115ef49060e076723cf784'),
    '4': ObjectId('6a115ef49060e076723cf785'),
    '5': ObjectId('6a115ef49060e076723cf786'),
    '6': ObjectId('6a115ef49060e076723cf787'),
    '7': ObjectId('6a115ef49060e076723cf788'),
    '8': ObjectId('6a115ef49060e076723cf789')
  }
}

```

```

> db.patients.aggregate([
  {$match:{treatmentCost:{$gt:20000}}},
  {$project:{_id:0,patientName:1,department:1}}
])
< {
  patientName: 'Meena',
  department: 'Neurology'
}
{
  patientName: 'Kiran',
  department: 'Neurology'
}
{
  patientName: 'Vijay',
  department: 'Cardiology'
}
{
  patientName: 'Priya',
  department: 'Neurology'
}

```

```
> db.patients.aggregate([
  {$sort:{treatmentCost:-1}},
  {$project:{_id:0,patientName:1,treatmentCost:1}}
])
< {
  patientName: 'Meena',
  treatmentCost: 30000
}
{
  patientName: 'Priya',
  treatmentCost: 27000
}
{
  patientName: 'Kiran',
  treatmentCost: 25000
}
{
  patientName: 'Vijay',
  treatmentCost: 22000
}
{
  patientName: 'Arun',
  treatmentCost: 20000
}
{
  patientName: 'Anil',
  treatmentCost: 18000
}
{
```

```
> db.patients.aggregate(  
  {$limit:5}  
)  
< {  
  _id: ObjectId('6a115ef49060e076723cf781'),  
  patientName: 'Arun',  
  doctor: 'Dr. Ravi',  
  department: 'Cardiology',  
  age: 45,  
  treatmentCost: 20000,  
  daysAdmitted: 5  
}  
{  
  _id: ObjectId('6a115ef49060e076723cf782'),  
  patientName: 'Meena',  
  doctor: 'Dr. Kumar',  
  department: 'Neurology',  
  age: 50,  
  treatmentCost: 30000,  
  daysAdmitted: 7  
}  
{  
  _id: ObjectId('6a115ef49060e076723cf783'),  
  patientName: 'Raj',  
  doctor: 'Dr. Ravi',  
  department: 'Cardiology',  
  age: 60,  
  treatmentCost: 15000,  
  daysAdmitted: 3  
}
```

```
> db.patients.aggregate(
  {$skip:3}
)
< {
  _id: ObjectId('6a115ef49060e076723cf784'),
  patientName: 'Divya',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 35,
  treatmentCost: 10000,
  daysAdmitted: 2
}
{
  _id: ObjectId('6a115ef49060e076723cf785'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
  treatmentCost: 25000,
  daysAdmitted: 6
}
{
  _id: ObjectId('6a115ef49060e076723cf786'),
  patientName: 'Sneha',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 30,
  treatmentCost: 8000,
  daysAdmitted: 2
}
```

```
> db.patients.aggregate([
  {$match:{department:"Cardiology"}},
  {$project:{_id:0,patientName:1}}
])
< {
  patientName: 'Arun'
}
{
  patientName: 'Raj'
}
{
  patientName: 'Vijay'
}
day2 >
```

```
> db.patients.aggregate(  
  {$group: {_id: "$department", totalAmount: {$sum: "$treatmentCost"}}}  
)  
< {  
  _id: 'Orthopedics',  
  totalAmount: 36000  
}  
{  
  _id: 'Cardiology',  
  totalAmount: 57000  
}  
{  
  _id: 'Neurology',  
  totalAmount: 82000  
}  
}
```

day2> |

```
> db.patients.aggregate([  
  {$group: {_id: "$doctor", totalAmount: {$avg: "$treatmentCost"}}},  
  {$sort: {totalAmount: 1}}  
)  
< {  
  _id: 'Dr. Mehta',  
  totalAmount: 12000  
}  
{  
  _id: 'Dr. Ravi',  
  totalAmount: 19000  
}  
{  
  _id: 'Dr. Kumar',  
  totalAmount: 27333.333333333332  
}  
}
```

```

> db.patients.aggregate([
  {$group: {_id: "$department", totalAmount: {$sum: "$treatmentCost"}}},
  {$match: {totalAmount: {$gt: 50000}}}
])
< {
  _id: 'Cardiology',
  totalAmount: 57000
}
{
  _id: 'Neurology',
  totalAmount: 82000
}
day2 >

```

```

> db.patients.aggregate([
  {$group: {_id: "$department", count: {$sum: 1}}},
  {$sort: {count: -1}}
])
< {
  _id: 'Orthopedics',
  count: 3
}
{
  _id: 'Cardiology',
  count: 3
}
{
  _id: 'Neurology',
  count: 3
}
day2 > |

```

```
> db.patients.aggregate([
  {$group: {_id: "$doctor", totalAmount: {$sum: "$treatmentCost"}}},
  {$sort: {totalAmount: -1}},
  {$limit: 3}
])
< {
  _id: 'Dr. Kumar',
  totalAmount: 82000
}
{
  _id: 'Dr. Ravi',
  totalAmount: 57000
}
{
  _id: 'Dr. Mehta',
  totalAmount: 36000
}
day2> |
```

```
> db.patients.aggregate([
  {$match: {age: {$gt: 40}}},
  {$group: {_id: "$department", Patientcount: {$sum: 1}}}
])
< {
  _id: 'Neurology',
  Patientcount: 2
}
{
  _id: 'Cardiology',
  Patientcount: 3
}
{
  _id: 'Orthopedics',
  Patientcount: 1
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", avgcost: {$avg: "$treatmentCost"}}},
  {$project: {_id: 0, departmentname: "$_id", avgcost: 1}},
  {$sort: {avgcost: -1}}
])
< {
  avgcost: 27333.333333333332,
  departmentname: 'Neurology'
}
{
  avgcost: 19000,
  departmentname: 'Cardiology'
}
{
  avgcost: 12000,
  departmentname: 'Orthopedics'
}
day2 >
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", patientcount: {$sum: 1}}},
  {$match: {patientcount: {$gt: 2}}}
])
< {
  _id: 'Neurology',
  patientcount: 3
}
{
  _id: 'Cardiology',
  patientcount: 4
}
{
  _id: 'Orthopedics',
  patientcount: 3
}
day2 >
```



```
> db.patients.aggregate([
  {$group: {_id: "$department", totalAmount: {$sum: "$treatmentCost"}}},
  {$sort: {totalAmount: -1}},
  {$skip: 1},
  {$limit: 3}
])
< {
  _id: 'Cardiology',
  totalAmount: 75000
}
{
  _id: 'Orthopedics',
  totalAmount: 36000
}
```

```
> db.patients.aggregate([
  {$match: {treatmentCost: {$gt: 10000}}},
  {$group: {_id: "$doctor", totalAmount: {$sum: "$treatmentCost"}}},
  {$sort: {totalAmount: -1}},
  {$limit: 2}
])
< {
  _id: 'Dr. Kumar',
  totalAmount: 82000
}
{
  _id: 'Dr. Ravi',
  totalAmount: 75000
}
day2> |
```